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HEAD CHIP, LIQUID JET HEAD, LIQUID JET RECORDING APPARATUS, AND METHOD OF MANUFACTURING HEAD CHIP

Abstract

A head chip, a liquid jet head, a liquid jet recording apparatus, and a method of manufacturing a head chip each capable of preventing an occurrence of an unwanted processing residual due to laser processing are provided. The head chip according to an aspect of the present disclosure includes a flow channel member which includes a recess formed of a first laser irradiation trace, and in which a liquid circulates in the recess, and a bonded member which is bonded to the flow channel member so as to close the recess. The flow channel member is provided with a second laser irradiation trace extending along an opening edge of the recess.

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Background/Summary

RELATED APPLICATIONS

[0001] This application claims priority to Japanese Patent application No. JP2024-026429, filed on Feb. 26, 2024, the entire content of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to a head chip, a liquid jet head, a liquid jet recording apparatus, and a method of manufacturing a head chip.

2. Description of the Related Art

[0003] A head chip to be installed in an inkjet printer may be provided with a flow channel member provided with a recess for ink to flow, and a bonded member which is bonded to the flow channel member so as to close the recess, in some cases. The recess is formed by, for example, laser processing (see, e.g., JP6-297180A (PTL1)).

[0004] In laser processing, a formation region of the recess out of a surface of the flow channel member is scanned with a laser beam. Thus, the recess due to the laser irradiation trace is formed in a scanning range with the laser beam.

[0005] However, when forming the recess with the laser processing, there is a possibility that a portion of the flow channel member melted with the laser irradiation flies to adhere to the vicinity of the opening edge of the recess as a foreign matter (hereinafter referred to as a “burr”). When an unwanted processing residual such as a burr protrudes from a surface of the flow channel member, there is a possibility that the unwanted processing residual is jammed between the flow channel member and the bonded member when bonding the bonded member to the surface of the flow channel member. When the unwanted processing residual is jammed, leakage of ink between the flow channel member and the bonded member, exfoliation of the bonded member, and so on are incurred.

[0006] The present disclosure provides a head chip, a liquid jet head, a liquid jet recording apparatus, and a method of manufacturing a head chip each capable of preventing an occurrence of the unwanted processing residual due to the laser processing.

SUMMARY OF THE INVENTION

[0007] In order to solve the problems described above, the present disclosure adopts the following aspects.

[0008] (1) A head chip according to an aspect of the present disclosure includes a flow channel member which includes a recess formed of a first laser irradiation trace, and in which a liquid circulates in the recess, and a bonded member which is bonded to the flow channel member so as to close the recess, wherein the flow channel member is provided with a second laser irradiation trace extending along an opening edge of the recess.

[0009] According to the present aspect, since the second laser irradiation trace is formed along the opening edge of the recess, the unwanted processing residuals remaining on the opening edge of the recess after forming the first laser irradiation trace can be reduced. Thus, it is possible to prevent the unwanted processing residuals from being jammed when bonding the flow channel

member and the bonded member to each other. As a result, the leakage of the liquid between the flow channel member and the bonded member is prevented, and at the same time, the bonding strength between the flow channel member and the bonded member can be ensured. Further, by reducing the unwanted processing residuals in the recess, it is possible to suppress the unevenness in the recess. Thus, it is possible to reduce the circulation resistance of the liquid which circulates in the recess to thereby smoothly circulate the liquid in the recess.

[0010] (2) In the head chip according to the aspect (1) described above, it is preferable that the recess includes a non-penetrating part configured to open only on a surface of the flow channel member, the surface facing to the bonded member side in a thickness direction, and a penetrating part which is configured to open on a bottom surface of the non-penetrating part, and penetrates the flow channel member in the thickness direction, and the second laser irradiation trace extends along at least an opening edge of the non-penetrating part.

[0011] According to the present aspect, there is a possibility that the unwanted processing residuals remaining on the opening edge of the non-penetrating part are exposed on the bonding surface of the flow channel member to the bonded member.

[0012] Therefore, by forming the second laser irradiation trace along at least the opening edge of the non-penetrating part, it is possible to prevent the unwanted processing residuals from being exposed on the bonding surface of the flow channel member to the bonded member.

[0013] (3) In the head chip according to one of the aspects (1) and (2) described above, it is preferable that the first laser irradiation trace extends in a first direction crossing a thickness direction of the flow channel member, and a plurality of first laser irradiation traces each identical to the first laser irradiation trace is formed in a second direction crossing the first direction when viewed from the thickness direction, and the second laser irradiation trace extends in the first direction along an outer side laser irradiation trace which is located at an outermost side in the second direction in the plurality of first laser irradiation traces.

[0014] According to the present aspect, by forming the second laser irradiation trace along the outer side laser irradiation trace out of the plurality of first laser irradiation traces, it is possible to effectively reduce the unwanted processing residuals remaining along the extending direction of the outer side laser irradiation trace.

[0015] (4) In the head chip according to the aspect (3) described above, it is preferable that a center in the second direction in the second laser irradiation trace is located at an inner side in the second direction from a center in the second direction in the outer side laser irradiation trace.

[0016] According to the present aspect, by disposing the center in the second direction in the second laser irradiation trace at an inner side from the center in the second direction in the outer side laser irradiation trace, it is possible to remove the unwanted processing residuals such as a burr due to an influence of the heat propagating to the periphery of the laser beam for forming the second laser irradiation trace. In this case, the influence of the heat propagating to the opening edge of the recess can be suppressed compared to when performing the scanning with the laser beam for forming the second laser irradiation trace on the same trajectory as the scan trajectory of the laser beam for forming the outer side laser irradiation trace. Thus, it is possible to prevent the unwanted processing residuals such as a wrinkle formed on the outer peripheral edge of the recess from being enlarged by the laser beam for forming the second laser irradiation trace. Therefore, it is possible to prevent the wrinkle and so on from being jammed as the unwanted processing residuals when bonding the flow channel member and the bonded member to each other. As a result, the leakage of the liquid between the flow channel member and the bonded member is prevented, and at the same time, the bonding strength between the flow channel member and the bonded member can be ensured.

[0017] (5) In the head chip according to any one of the aspects (1) through (4) described above, it is preferable that the recess is formed to have a rectangular shape having a longitudinal direction set to the first direction and a transverse direction set to the second direction when viewed from the

thickness direction.

[0018] According to the present aspect, since the longitudinal direction of the recess and the extending direction of the second laser irradiation trace coincide with each other, the unwanted processing residuals can be reduced along the longitudinal direction of the recess.

[0019] (6) In the head chip according to any one of the aspects (1) through (5) described above, it is preferable that the recess includes a non-penetrating part configured to open only on a surface of the flow channel member, the surface facing to the bonded member side in a thickness direction, and a penetrating part which is configured to open on a bottom surface of the non-penetrating part, and penetrates the flow channel member in the thickness direction, and the penetrating part includes a first penetrating part, and a second penetrating part which is disposed at a distance in the first direction with respect to the first penetrating part, and is communicated with the first penetrating part through the non-penetrating part, a chip main body provided with a pressure chamber communicated with the recess through the first penetrating part and a circulation channel communicated with the recess through the second penetrating part is bonded to the flow channel member at an opposite side to the bonded member in the thickness direction, and the bonded member is provided with a jet hole configured to jet a liquid in the recess to an outside.

[0020] According to the present aspect, the flow channel member has a function as a return plate which is communicated with the jet hole, and circulates the liquid to a jet channel and a circulation channel. In this case, in the head chip according to the present disclosure, since the unevenness in the recess can be suppressed, the circulation of the liquid between the jet hole, the jet channel, and the circulation channel through the recess can smoothly be performed.

[0021] (7) A liquid jet head according to an aspect of the present disclosure preferably includes the head chip according to any one of the aspects (1) through (6) described above.

[0022] According to the present aspect, it is possible to provide a liquid jet head excellent in reliability.

[0023] (8) A liquid jet recording apparatus according to an aspect of the present disclosure includes the liquid jet head according to the aspect (7) described above.

[0024] According to the present aspect, it is possible to provide a liquid jet recording apparatus excellent in reliability.

[0025] (9) A method of manufacturing a head chip according to an aspect of the present disclosure is a method of manufacturing a head chip including a flow channel member provided with a recess through which a liquid circulates, and a bonded member which is bonded to the flow channel member so as to close the recess, the method including a first laser irradiation step of irradiating a bonding surface of the flow channel member to the bonded member with a first laser beam to thereby form an outer shape of the recess with a first laser irradiation trace, and a second laser irradiation step of irradiating the bonding surface with a second laser beam along an opening edge of the recess to thereby form a second laser irradiation trace along the opening edge of the recess.

[0026] According to an aspect of the present disclosure, it is possible to prevent the unwanted processing residuals due to the laser processing from occurring.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 is a schematic diagram of a printer according to a first embodiment.

[0028] FIG. 2 is a schematic configuration diagram of an inkjet head and an ink circulation mechanism related to the first embodiment.

[0029] FIG. 3 is an exploded perspective view of a head chip according to the first embodiment.

[0030] FIG. 4 is a cross-sectional view along the line IV-IV shown in FIG. 3.

[0031] FIG. 5 is a cross-sectional view along the line V-V shown in FIG. 3.

[0032] FIG. **6** is a cross-sectional view corresponding to the line VI-VI shown in FIG. **4**.
[0033] FIG. **7** is an enlarged view of the VII portion shown in FIG. **4**.
[0034] FIG. **8** is a cross-sectional view corresponding to the line VIII-VIII shown in FIG. **4**.
[0035] FIG. **9** is a flowchart illustrating a method of manufacturing the head chip according to the first embodiment.
[0036] FIG. **10** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0037] FIG. **11** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0038] FIG. **12** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0039] FIG. **13** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0040] FIG. **14** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0041] FIG. **15** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0042] FIG. **16** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0043] FIG. **17** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0044] FIG. **18** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0045] FIG. **19** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0046] FIG. **20** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0047] FIG. **21** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0048] FIG. **22** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0049] FIG. **23** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0050] FIG. **24** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0051] FIG. **25** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0052] FIG. **26** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0053] FIG. **27** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0054] FIG. **28** is a process diagram illustrating the method of manufacturing the head chip according to the first embodiment.
[0055] FIG. **29** is a process diagram illustrating a method of manufacturing a head chip according to a second embodiment.
[0056] FIG. **30** is a process diagram illustrating the method of manufacturing the head chip according to the second embodiment.
[0057] FIG. **31** is a process diagram illustrating the method of manufacturing the head chip according to the second embodiment.
[0058] FIG. **32** is a process diagram illustrating the method of manufacturing the head chip

according to the second embodiment.

[0059] FIG. **33** is a flowchart illustrating a method of manufacturing a head chip according to a third embodiment.

[0060] FIG. **34** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0061] FIG. **35** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0062] FIG. **36** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0063] FIG. **37** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0064] FIG. **38** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0065] FIG. **39** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0066] FIG. **40** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0067] FIG. **41** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0068] FIG. **42** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0069] FIG. **43** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0070] FIG. **44** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0071] FIG. **45** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0072] FIG. **46** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0073] FIG. **47** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0074] FIG. **48** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0075] FIG. **49** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0076] FIG. **50** is a process diagram illustrating the method of manufacturing the head chip according to the third embodiment.

[0077] FIG. **51** is a process diagram illustrating a method of manufacturing a head chip according to a fourth embodiment.

[0078] FIG. **52** is a process diagram illustrating the method of manufacturing the head chip according to the fourth embodiment.

[0079] FIG. **53** is a process diagram illustrating the method of manufacturing the head chip according to the fourth embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0080] Some embodiments according to the present disclosure will hereinafter be described with reference to the drawings. In the embodiments and modified examples hereinafter described, constituents corresponding to each other will be denoted by the same reference symbols to omit the descriptions thereof in some cases. In the following descriptions, expressions representing relative or absolute arrangements such as “parallel,” “perpendicular,” “central,” and “coaxial” not only represent strictly such arrangements, but also represent the state of being relatively displaced with a

tolerance, or an angle or a distance to the extent that the same function can be obtained. In the following embodiment, the description will be presented citing an inkjet printer (hereinafter referred to simply as a printer) for performing recording on a recording target medium using ink (a liquid) as an example. The scale size of each member is arbitrarily modified so as to provide a recognizable size to the member in the drawings used in the following description.

First Embodiment

[Printer 1]

[0081] FIG. 1 is a schematic configuration diagram of a printer 1.

[0082] As shown in FIG. 1, the printer (a liquid jet recording device) 1 according to the first embodiment is provided with a pair of conveyance mechanisms 2, 3, ink tanks 4, inkjet heads (liquid jet heads) 5, ink circulation mechanisms 6, and a scanning mechanism 7.

[0083] In the following explanation, the description is presented using an orthogonal coordinate system of X, Y, and Z as needed. In this case, the X direction coincides with a conveying direction (a sub-scanning direction) of a recording target medium P (e.g., paper). The Y direction coincides with a scanning direction (a main scanning direction) of the scanning mechanism 7. The Z direction represents a height direction (a gravitational direction) perpendicular to the X direction and the Y direction. In the following explanation, the description will be presented defining an arrow side as a positive (+) side, and an opposite side to the arrow as a negative (−) side in the drawings in each of the X direction, the Y direction, and the Z direction. In the first embodiment, the +Z side corresponds to an upper side in the gravitational direction, and the −Z side corresponds to a lower side in the gravitational direction.

[0084] The conveyance mechanisms 2, 3 convey the recording target medium P toward the +X side. The conveyance mechanisms 2, 3 each include a pair of rollers 11, 12 extending in, for example, the Y direction.

[0085] The ink tanks 4 respectively contain ink of four colors such as yellow, magenta, cyan, and black. The inkjet heads 5 are configured so as to be able to respectively eject the four colors of ink, namely the yellow ink, the magenta ink, the cyan ink, and the black ink according to the ink tanks 4 coupled thereto.

[0086] FIG. 2 is a schematic configuration diagram of the inkjet head 5 and the ink circulation mechanism 6.

[0087] As shown in FIGS. 1, 2, the ink circulation mechanism 6 circulates the ink between the ink tank 4 and the inkjet head 5. Specifically, the ink circulation mechanism 6 is provided with a circulation flow channel 23 having an ink supply tube 21 and an ink discharge tube 22, a pressure pump 24 coupled to the ink supply tube 21, and a suction pump 25 coupled to the ink discharge tube 22.

[0088] The pressure pump 24 pressurizes an inside of the ink supply tube 21 to deliver the ink to the inkjet head 5 through the ink supply tube 21. Thus, the ink supply tube 21 side is provided with positive pressure with respect to the inkjet head 5.

[0089] The suction pump 25 depressurizes the inside of the ink discharge tube 22 to suction the ink from the inkjet head 5 through the ink discharge tube 22. Thus, the ink discharge tube 22 side is provided with negative pressure with respect to the inkjet head 5. It is arranged that the ink can circulate between the inkjet head 5 and the ink tank 4 through the circulation flow channel 23 by driving the pressure pump 24 and the suction pump 25.

[0090] The scanning mechanism 7 makes the inkjet heads 5 perform a reciprocal scan in the Y direction. The scanning mechanism 7 is provided with a guide rail 28 extending in the Y direction, and a carriage 29 movably supported by the guide rail 28.

<Inkjet Heads 5>

[0091] As shown in FIG. 1, the inkjet heads 5 are mounted on the carriage 29. In the illustrated example, the plurality of inkjet heads 5 is mounted on the single carriage 29 so as to be arranged side by side in the Y direction. The inkjet heads 5 are each provided with a head chip 50 (see FIG.

3), an ink supply unit (not shown) for coupling the ink circulation mechanism 6 and the head chip 50, and a control unit (not shown) for applying drive voltages to the head chip 50.

<Head Chip 50>

[0092] FIG. 3 is an exploded perspective view of the head chip 50. FIG. 4 is a cross-sectional view along the line IV-IV shown in FIG. 3. FIG. 5 is a cross-sectional view along the line V-V shown in FIG. 3.

[0093] As shown in FIG. 3 through FIG. 5, the head chip 50 is of a circulation type (a vertical circulation type) which circulates the ink with the ink tank 4 out of so-called edge-shoot types which eject the ink from a tip portion in a channel extension direction (the Z direction) in each of ejection channels 71 described later.

[0094] The head chip 50 is provided with a first chip module 51A, a second chip module 51B, a return plate 52, and a nozzle plate 53. In the following explanation, a configuration of each of the chip modules 51A, 51B will be described citing the first chip module 51A as an example.

Therefore, the constituents in the second chip module 51B substantially the same as those of the first chip module 51A are denoted by the same reference symbols as in the first chip module 51A, and the description thereof will be omitted in some cases.

<First Chip Module 51A>

[0095] The first chip module 51A is provided with a first actuator plate 61, a first cover plate 62, and a first back plate 63. In the following explanation, the first chip module 51A will be described defining the +Y side as an obverse surface side, and the -Y side as a reverse surface side.

[0096] The first actuator plate 61 is formed of a laminated substrate (a so-called chevron type) having two piezoelectric substrates which are different in polarization direction along the thickness direction (the Y direction) from each other, and are stacked on one another. It should be noted that as the piezoelectric substrates, there is preferably used a ceramics substrate formed of, for example, PZT (lead zirconate titanate). It should be noted that the first actuator plate 61 can be formed of a single piezoelectric substrate in which the polarization direction is set in a single direction (a so-called monopole type).

[0097] The first actuator plate 61 is provided with the ejection channels 71 each filled with the ink, and non-ejection channels 72 not filled with the ink. The channels 71, 72 are alternately arranged at intervals in the X direction in the first actuator plate 61 to thereby form a channel array 70. The configuration in which the channel extension direction coincides with the Z direction (an extension direction) will be described in the first embodiment, but the channel extension direction can cross the Z direction.

[0098] As shown in FIG. 3 and FIG. 4, the ejection channels 71 each have an upper end portion terminating within the first actuator plate 61, and a lower end portion opening on a lower end surface of the first actuator plate 61. Meanwhile, an upper part of each of the ejection channels 71 gradually shallows in depth in the Y direction along the upward direction. A lower part of each of the ejection channels 71 penetrates the first actuator plate 61 in the Y direction.

[0099] As shown in FIG. 3 and FIG. 5, the non-ejection channel 72 penetrates the first actuator plate 61 in the Z direction. The non-ejection channel 72 penetrates the first actuator plate 61 in the Y direction over the entire length in the Z direction. In other words, the depth in the Y direction in the non-ejection channel 72 is uniform throughout the entire length in the Z direction.

[0100] In the first actuator plate 61, a portion located between each of the ejection channels 71 and corresponding one of the non-ejection channels 72 constitutes a drive wall 75. Therefore, both sides in the X direction of the ejection channel 71 are surrounded by the pair of drive walls 75. In the first actuator plate 61, a portion located above the ejection channel 71 constitutes a tail part 76.

<First Cover Plate 62>

[0101] As shown in FIG. 3 through FIG. 5, the first cover plate 62 is bonded to the obverse surface of the first actuator plate 61. Specifically, the first cover plate 62 closes the obverse surface-side openings of the channels 71, 72 in a state of exposing the obverse surfaces of the tail parts 76. A

lower end surface of the first cover plate **62** is arranged so as to be coplanar with the lower end surface of the first actuator plate **61**.

[0102] In the first cover plate **62**, at positions overlapping the upper parts of the ejection channels **71** when viewed from the Y direction, there is formed a common ink chamber **80**. The common ink chamber **80** extends in the X direction with a length sufficient for straddling, for example, the channel array **70**, and at the same time, opens on the obverse surface of the first cover plate **62**. The common ink chamber **80** is indirectly connected to the ink supply tube **21** through an entrance port not shown.

[0103] In the common ink chamber **80**, at the positions overlapping the upper parts of the respective ejection channels **71** when viewed from the Y direction, there are individually formed slits **81**. The slits **81** each communicate the inside of the ejection channel **71** and the inside of the common ink chamber **80** with each other through the upper part of the ejection channel **71**. Therefore, the common ink chamber **80** is communicated with the ejection channels **71** through the respective slits **81** on the one hand, but is not communicated with the non-ejection channels **72** on the other hand.

<First Back Plate **63**>

[0104] The first back plate **63** is bonded to a reverse surface of the first actuator plate **61**. The first back plate **63** has an equivalent outer shape to that of the first actuator plate **61** when viewed from the Y direction. The first back plate **63** is overlapped with the whole of the first actuator plate **61** when viewed from the Y direction. In other words, the first back plate **63** closes reverse surface side openings of the channels **71**, **72**.

[0105] As shown in FIG. **3**, the first actuator plate **61** is provided with common wiring lines **85** and individual wiring lines **86** as drive wiring lines.

[0106] As shown in FIG. **3** and FIG. **4**, the common wiring lines **85** are each provided with a common electrode **87** and a common terminal **88**.

[0107] The common electrode **87** is formed on inner side surfaces opposed to each other in the X direction out of the inner surfaces of the ejection channel **71**. In the illustrated example, the common electrode **87** is formed throughout the entire area in the Y direction and the Z direction on the inner side surfaces of the ejection channel **71**. It should be noted that it is sufficient for the common electrode **87** to be provided to at least a portion located below the lower-end opening edge of the slit **81** out of the inner side surface of the ejection channel **71**.

[0108] The common terminal **88** is formed on an obverse surface of the tail part **76**. The common terminal **88** is disposed on the obverse surface of the tail part **76** so as to correspond to each of the ejection channels **71**. Each of the common terminals **88** extends linearly in the Z direction above corresponding one of the ejection channels **71**. The lower end portion in the common terminal **88** is connected to the common electrode **87** in an upper-end opening edge of the ejection channel **71**.

[0109] As shown in FIG. **3** and FIG. **5**, the individual wiring lines **86** are each provided with individual electrodes **91**, and an individual terminal **92**.

[0110] The individual electrodes **91** are each formed on one of the inner side surfaces opposed to each other in the X direction out of the inner surfaces of each of the non-ejection channels **72**. In the illustrated example, the individual electrode **91** is formed throughout the entire area in the Y direction and the Z direction on the inner side surface of the non-ejection channel **72**. It should be noted that it is sufficient for the individual electrode **91** to be formed at a position opposed to the common electrode **87** in at least a portion located below the lower-end opening edge of the slit **81**.

[0111] The individual terminal **92** is provided to a portion located above the common terminal **88** on the obverse surface of the tail part **76**. The individual terminal **92** is formed to have a strip shape extending in the X direction. The individual terminal **92** couples the individual electrodes **91**, which are opposed to each other in the X direction across the ejection channel **71**, to each other at obverse surface side opening edges of the non-ejection channels **72**, which are opposed to each other in the X direction across the ejection channel **71**. In the tail part **76**, in a portion located

between the common terminal **88** and the individual terminal **92**, there is formed a dividing groove **79**. The dividing groove **79** opens on the obverse surface of the tail part **76**, and at the same time, extends in the X direction. The dividing groove **79** separates the common terminal **88** and the individual terminal **92** from each other. The dividing groove **79** is formed to have a depth in the Y direction smaller than a half of the thickness of the first actuator plate **61**. It should be noted that the depth of the dividing groove **79** can be changed as appropriate within a range in which the individual electrode **91** is not cut.

[0112] To the obverse surface of the tail part **76**, there is pressure-bonded a flexible printed board (not shown). The flexible printed board is coupled to the common terminals **88** and the individual terminals **92** on the obverse surfaces of the tail parts **76**. The flexible printed board couples the first chip module **51A** and the control unit to each other.

<Second Chip Module **51B**>

[0113] The second chip module **51B** is provided with a second actuator plate **101**, a second cover plate **102**, and a second back plate **103**. The second chip module **51B** has the second back plate **103**, the second actuator plate **101**, and the second cover plate **102** overlapped in sequence from the +Y side toward the -Y side. The second chip module **51B** is overlapped with the first chip module **51A** in a state in which the obverse surface side (the -Y side) faces to an opposite side to the first chip module **51A**. Specifically, the first chip module **51A** and the second chip module **51B** are integrated with each other by the reverse surfaces of the first back plate **63** and the second back plate **103** being bonded to each other. In this case, the lower end surfaces of the respective chip modules **51A**, **51B** are arranged so as to be coplanar with each other.

[0114] The ejection channels **71** and the non-ejection channels **72** of the second chip module **51B** are arranged so as to be shifted as much as a half pitch with respect to the arrangement pitch of the ejection channels **71** and the non-ejection channels **72** of the first chip module **51A**. In other words, the ejection channels **71** of the chip modules **51A**, **51B**, and the non-ejection channels **72** of the chip modules **51A**, **51B** are each arranged in a zigzag manner. In this case, the ejection channels **71** of the first chip module **51A** and the non-ejection channels **72** of the second chip module **51B** face each other in the Y direction, and the non-ejection channels **72** of the first chip module **51A** and the ejection channels **71** of the second chip module **51B** face each other in the Y direction. It should be noted that the pitch of the channels **71**, **72** in each of the chip modules **51A**, **51B** can be changed as appropriate.

<Return Plate **52**>

[0115] The return plate **52** is bonded to the lower end surfaces of the respective chip modules **51A**, **51B** in a lump via an adhesive. The return plate **52** closes the lower end opening parts of the respective channels **71**, **72**. The return plate **52** is formed of, for example, polyimide. The return plate **52** is provided with a plurality of first communication channels **110** and a plurality of second communication channels **111**. It should be noted that the first communication channels **110** and the second communication channels **111** are formed of a plurality of lines of laser irradiation traces as described later in the manufacturing method. It should be noted that in FIG. 4 to FIG. 7, inner surface shapes of the first communication channels **110** and the second communication channels **111** are shown as smooth surfaces for the sake of convenience.

[0116] FIG. 6 is a cross-sectional view corresponding to the line VI-VI shown in FIG. 4. FIG. 7 is an enlarged view of the VII portion shown in FIG. 4.

[0117] As shown in FIG. 6 and FIG. 7, the plurality of first communication channels **110** is formed individually at equivalent positions in the X direction to those of the respective ejection channels **71** in the first chip module **51A**. In the first embodiment, the plurality of first communication channels **110** is formed at intervals in the X direction so as to correspond to the arrangement pitch of the ejection channels **71**. Each of the first communication channels **110** is a recess formed to have a U-shape in a side view viewed from the X direction. Specifically, each of the first communication channels **110** is provided with an upstream opening (penetrating part) **115**, a

downstream opening (penetrating part) **116**, and a connecting part (non-penetrating part) **117**. Since the first communication channels **110** have substantially the same configurations, in the following configuration, the details of the first communication channels **110** will be described citing one of the first communication channels **110** as an example.

[0118] The upstream opening **115** is formed at a position overlapping the ejection channel **71** when viewed from the Z direction. The upstream opening **115** has an upper end portion opening on an upper surface of the return plate **52**, and a lower end portion terminating in the return plate **52**. The upstream opening **115** is communicated with the inside of the ejection channel **71** through the lower-end opening of the ejection channel **71**.

[0119] At least a part of an opening edge of the upstream opening **115** is arranged at an outer side of a lower-end opening edge of the ejection channel **71** when viewed from the Z direction. Specifically, a dimension in the X direction in the upstream opening **115** is smaller than a dimension in the X direction in the lower end opening of the ejection channel **71**. A dimension in the Y direction in the upstream opening **115** is larger than a dimension in the Y direction in the lower end opening of the ejection channel **71**. In the lower end surface of the first chip module **51A**, portions located at both sides in the Y direction with respect to the ejection channel **71** are exposed through the upstream opening **115**.

[0120] The downstream opening **116** is formed at a position overlapping the first back plate **63** when viewed from the Z direction. The downstream opening **116** has an upper end portion opening on the upper surface of the return plate **52**, and a lower end portion terminating in the return plate **52**. A dimension in the Z direction in the downstream opening **116** is made equivalent to that of the upstream opening **115**.

[0121] The connecting part **117** opens only on the lower surface of the return plate **52**, and at the same time, communicates the upstream opening **115** and the downstream opening **116** with each other. The connecting part **117** is formed to have a rectangular shape having a transverse direction set to the X direction and a longitudinal direction set to the Y direction when viewed from the Z direction. In the first embodiment, the dimension in the Z direction in the connecting part **117** is larger than the dimensions in the Z direction in the upstream opening **115** and the downstream opening **116**. It should be noted that the dimension in the Z direction in the connecting part **117** may be made different in accordance with a position in the Y direction.

[0122] The connecting part **117** is larger in dimension in the X direction than the upstream opening **115** and the downstream opening **116** when viewed from the Z direction. Specifically, the connecting part **117** is provided with an upstream wide part **117a** and a downstream wide part **117b**.

[0123] The upstream wide part **117a** is arranged at a position overlapping the upstream opening **115** when viewed from the Z direction. The upstream wide part **117a** is made one size larger than the upstream opening **115** when viewed from the Z direction. The upstream opening **115** opens on a surface (a bottom surface) facing downward out of the inner surfaces of the upstream wide part **117a**. Therefore, the upstream opening **115** penetrates the return plate **52** in the Z direction via the upstream wide part **117a**.

[0124] The downstream wide part **117b** extends toward the -Y side from the upstream wide part **117a**. The downstream wide part **117b** overlaps the downstream opening **116** when viewed from the Z direction. The downstream wide part **117b** is made one size larger than the downstream opening **116** when viewed from the Z direction. A dimension in the X direction in the downstream wide part **117b** is made smaller than that of the upstream wide part **117a**. The downstream opening **116** opens on the bottom surface of the downstream wide part **117b**. Therefore, the downstream opening **116** penetrates the return plate **52** in the Z direction via the downstream wide part **117b**.

[0125] As shown in FIG. 5, the plurality of second communication channels **111** is formed individually at equivalent positions in the X direction to those of the respective ejection channels **71** in the second chip module **51B**. In the first embodiment, the plurality of second communication channels **111** is formed at intervals in the X direction so as to correspond to the arrangement pitch

of the ejection channels **71** of the second chip module **51B**. Specifically, the first communication channels **110** and the second communication channels **111** are alternately arranged at intervals in the X direction. It should be noted that the second communication channels **111** have substantially the same configuration as that of the first communication channels **110**. Therefore, substantially the same constituents of the second communication channels **111** as those of the first communication channels **110** are denoted by the same reference symbols to omit the detailed description of the second communication channels **111**.

[0126] FIG. **8** is a cross-sectional view corresponding to the line VIII-VIII shown in FIG. **4**.

[0127] As shown in FIG. **4** and FIG. **8**, the first back plate **63** and the second back plate **103** constitute a flow channel plate **120** in a state in which the first back plate **63** and the second back plate **103** overlap each other. The flow channel plate **120** is provided with a plurality of first connecting channels **121**, a plurality of second connecting channels **122**, and a manifold **123**.

[0128] The plurality of first connecting channels **121** is formed individually at positions overlapping the downstream openings **116** of the respective first communication channels **110** when viewed from the Z direction. The first connecting channels **121** are arranged at intervals in the X direction at the same pitch as that of the first communication channels **110**. Specifically, the first connecting channels **121** open on the reverse surface of the first back plate **63**. The reverse surface side openings in the first connecting channels **121** are closed by the second back plate **103**.

[0129] The first connecting channels **121** each extend linearly in the Z direction when viewed from the Y direction. A lower end portion in each of the first connecting channels **121** opens on the lower end surface of the first back plate **63**. Thus, the lower end opening of each of the first connecting channels **121** is communicated with the downstream opening **116**. In contrast, an upper end portion in each of the first connecting channels **121** terminates in the first back plate **63**.

[0130] As shown in FIG. **6**, the lower end opening of each of the first connecting channels **121** is made larger in dimension in the X direction than the downstream opening **116**, and is made smaller in dimension in the X direction than the downstream wide part **117b**. In this case, an opening edge of the downstream opening **116** flares to the inside of the lower-end opening edge of the first connecting channel **121**. However, it is possible for the lower end opening in each of the first connecting channels **121** to be made larger in dimension in the X direction than the downstream wide part **117b**, or made smaller in dimension in the X direction than the downstream opening **116**.

[0131] As shown in FIG. **8**, the plurality of second connecting channels **122** is formed individually at positions overlapping the downstream openings **116** of the respective second communication channels **111** when viewed from the Z direction. The second connecting channels **122** are arranged at intervals in the X direction at the same pitch as that of the second communication channels **111**. Specifically, the first connecting channels **121** and the second connecting channels **122** are alternately arranged in the X direction.

[0132] As shown in FIG. **5** and FIG. **6**, each of the second connecting channels **122** is communicated with the downstream opening **116** of corresponding one of the second communication channels **111**. Specifically, the second connecting channels **122** open on the reverse surface (a surface facing to the +Y side) of the second back plate **103**. The reverse surface side openings in the second connecting channels **122** are closed by the first back plate **63**. The second connecting channels **122** extend in the Z direction. A lower end portion in each of the second connecting channels **122** opens on the lower end surface of the second back plate **103**. Thus, the lower end opening of each of the second connecting channels **122** is communicated with the downstream opening **116** of corresponding one of the second communication channels **111**. In contrast, an upper end portion in each of the second connecting channels **122** terminates in the second back plate **103**. It should be noted that it is possible to set the dimensions and so on of the second connecting channels **122** to substantially the same as those of the first connecting channels **121**.

[0133] The manifold **123** is provided to a portion located above the connecting channels **121**, **122**

in the flow channel plate **120**. The manifold **123** is formed by overlapping a first recess **123a** provided to the first back plate **63** and a second recess **123b** provided to the second back plate **103** each other. The first recess **123a** is a recess which opens on the reverse surface of the first back plate **63**, and which extends in the Z direction and the Y direction. The second recess **123b** is a recess which opens on the reverse surface of the second back plate **103**, and which extends in the Z direction and the Y direction. The manifold **123** is formed by communicating the reverse surface side openings of the first recess **123a** and the second recess **123b** with each other. It should be noted that the manifold **123** may be provided with a configuration in which a recess provided to either one of the first back plate **63** and the second back plate **103** is closed by the reverse surface of the other of the back plates.

[0134] The connecting channels **121**, **122** are communicated in a lump with the manifold **123**. Specifically, the upper end opening of each of the first connecting channels **121** opens on the lower end surface of the first recess **123a**. An upper end opening of each of the second connecting channels **122** opens on the lower end surface of the second recess **123b**. It should be noted that the manifold **123** is indirectly connected to the ink discharge tube **22** through an exit port not shown.

[0135] The chip modules **51A**, **51B** and the return plate **52** are covered with protective films **125**. In the first embodiment, the protective films **125** are also formed on an inner surface of the common ink chamber **80**, inner surfaces of the slits **81**, inner surfaces of the ejection channels **71**, inner surfaces of the communication channels **110**, **111**, inner surfaces of the connecting channels **121**, **122**, and an inner surface of the manifold **123**. The protective films **125** each include an organic insulating material such as a para-xylylene resin material (e.g., parylene (a registered trademark)) as a material having an insulating property. The protective films **125** can be formed of tantalum oxide (Ta.sub.2O.sub.5), silicon nitride (SiN), silicon carbide (SiC), silicon oxide (SiO.sub.2), diamond-like carbon, or the like, or can include at least any one of these materials.

<Nozzle Plate **53**>

[0136] As shown in FIG. **3** through FIG. **5**, the nozzle plate **53** is bonded to the lower end surface of the return plate **52**. A plurality of nozzle holes (first nozzle holes **131** and second nozzle holes **132**) each penetrating the nozzle plate **53** in the Z direction is arranged in the nozzle plate **53**.

[0137] In the nozzle plate **53**, the plurality of first nozzle holes **131** is formed individually at positions overlapping the respective first communication channels **110** when viewed from the Z direction. In other words, the first nozzle holes **131** are arranged at intervals in the X direction at the same pitch as that of the first communication channels **110**. The first nozzle holes **131** are communicated with the corresponding ejection channels **71** of the first chip module **51A** through the corresponding first communication channels **110**, respectively. Specifically, the first nozzle holes **131** are each formed at a position overlapping the ejection channel **71** and the upstream wide part **117a** when viewed from the Z direction in a +Y-side end portion in corresponding one of the first communication channels **110**. It should be noted that the first nozzle holes **131** can be communicated with the first communication channels **110** at positions shifted in the Y direction from the ejection channels **71** of the first chip module **51A**, respectively.

[0138] In the nozzle plate **53**, the plurality of second nozzle holes **132** is formed individually at positions overlapping the respective second communication channels **111** when viewed from the Z direction. In other words, the second nozzle holes **132** are arranged at intervals in the X direction at the same pitch as that of the second communication channels **111**. The second nozzle holes **132** are communicated with the corresponding ejection channels **71** of the second chip module **51B** through the corresponding second communication channels **111**, respectively. Specifically, the second nozzle holes **132** are each formed at a position overlapping the ejection channel **71** and the upstream wide part **117a** when viewed from the Z direction in a -Y-side end portion in corresponding one of the second communication channels **111**. It should be noted that the second nozzle holes **132** may be communicated with the second communication channels **111** at positions shifted in the Y direction from the ejection channels **71** of the second chip module **51B**,

respectively.

[Operation Method of Printer 1]

[0139] Then, there will hereinafter be described when recording a character, a figure, or the like on the recording target medium P using the printer 1 configured as described above.

[0140] It should be noted that it is assumed that as an initial state, the sufficient ink having colors different from each other is respectively encapsulated in the four ink tanks 4 shown in FIG. 1.

Further, there is provided a state in which the inkjet heads 5 are filled with the ink in the ink tanks 4 via the ink circulation mechanisms 6, respectively.

[0141] Under such an initial state, when making the printer 1 operate, the recording target medium P is conveyed toward the +X side while being pinched by the rollers 11, 12. By the carriage 29 moving in the Y direction at the same time as the conveyance of the recording target medium P, the inkjet heads 5 mounted on the carriage 29 make a reciprocal motion in the Y direction.

[0142] Here, the operation of each of the inkjet heads 5 will hereinafter be described in detail.

[0143] In such a vertically circulating type head chip 50 as in the first embodiment, first, by making the pressure pump 24 and the suction pump 25 shown in FIG. 2 operate, the ink is circulated in the circulation flow channel 23. In this case, the ink circulating through the ink supply tube 21 flows into the common ink chamber 80 of each of chip modules 51 through the entrance port. The ink having flowed into the common ink chambers 80 is supplied to the inside of each of the ejection channels through the slit 81. The ink having flowed into the ejection channels 71 gathers in the manifold 123 through the communication channels 110, 111 and the connecting channels 121, 122, and is then discharged to the ink discharge tube 22 through the exit port. The ink discharged to the ink discharge tube 22 is returned to the ink tank 4, and is then supplied again to the ink supply tube 21. Thus, the ink is circulated between the inkjet head 5 and the ink tank 4.

[0144] Then, when the reciprocation is started by the carriage 29, the drive voltages are applied to the electrodes 87, 91 via the flexible boards. On this occasion, the drive voltage is applied between the electrodes 87, 91 by setting the individual electrode 91 at a drive potential Vdd, and the common electrode 87 at a reference potential GND. Then, a thickness shear deformation occurs in the two drive walls 75 partitioning the ejection channel 71, and the two drive walls 75 each deform so as to protrude toward the non-ejection channel 72. Specifically, the actuator plates 61, 101 in the first embodiment each have two piezoelectric substrates on which the polarization treatment has been performed in the thickness direction (the Y direction), and which are stacked on one another, and therefore, by applying the drive voltage, the actuator plates 61, 101 each make a flexural deformation having a V-shape centering on an intermediate position in the Y direction in the drive walls 75. Thus, the ejection channel 71 deforms as if it bulges.

[0145] When the volume of the ejection channel 71 increases due to the deformation of the two drive walls 75, the ink in the common ink chamber 80 is induced into the ejection channel 71 through the slit 81. Then, the ink induced to the inside of the ejection channel 71 propagates to the inside of the ejection channel 71 as a pressure wave, and the drive voltage applied between the electrodes 87, 91 is set to zero at the timing at which the pressure wave reaches the nozzle hole 131, 132.

[0146] Thus, the drive walls 75 are restored, and the volume of the ejection channel 71 having once increased is restored to the original volume. Due to this operation, the internal pressure of the ejection channel 71 increases to pressurize the ink. As a result, it is possible to eject the ink from the nozzle hole 131, 132. On this occasion, the ink turns to an ink droplet having a droplet shape when passing through the nozzle hole 131, 132, and is then ejected. Thus, it is possible to record a character, an image, or the like on the recording target medium P as described above. In other words, in the head chip 50 according to the first embodiment, out of the ink flowing through each of the communication channels 110, 111, a part is ejected through corresponding one of the nozzle holes 131, 132, while the rest is returned to the manifold 123 through corresponding one of the connecting channels 121, 122.

[Method of Manufacturing Head Chip 50]

[0147] Then, a method of manufacturing the head chip **50** described above will be described. FIG. **9** is a flowchart illustrating the method of manufacturing the head chip **50**. In the following description, there is described when manufacturing the head chip **50** chip by chip as an example for the sake of convenience.

[0148] As shown in FIG. **9**, the method of manufacturing the head chip **50** is provided with a module forming step **S1**, a module stacking step **S2**, a return plate stacking step **S3**, a return plate processing step **S4**, a protective film forming step **S5**, and a nozzle plate stacking step **S6**.

[0149] In the module forming step **S1**, each of the first chip module **51A** and the second chip module **51B** is formed.

[0150] FIG. **10** through FIG. **12** are each a process diagram illustrating the method of manufacturing the head chip **50**, and are each a cross-sectional view corresponding to FIG. **4**.

[0151] As shown in FIG. **10**, in the module stacking step **S2**, the chip modules **51A**, **51B** formed in the module forming step **S1** are bonded to each other. Specifically, the reverse surfaces of the back plates **63**, **103** are bonded to each other in a state in which the lower end surfaces of the respective chip modules **51A**, **51B** coincide with each other. Thus, the reverse side openings of the first connecting channels **121** are closed by the second back plate **103**, the reverse side openings of the second connecting channels **122** are closed by the first back plate **63**, and at the same time, the manifold **123** is formed with the first recess **123a** and the second recess **123b**. Thus, the stacked body of the chip modules **51A**, **51B** is formed.

[0152] As shown in FIG. **11**, in the return plate stacking step **S3**, the return plate **52** is bonded to a lower end surface in the stacked body of the chip modules **51A**, **51B**.

[0153] As shown in FIG. **12**, in the return plate processing step **S4**, the communication channels **110**, **111** are provided to portions of the return plate **52** overlapping the ejection channels **71** when viewed from the Z direction. The communication channels **110**, **111** are formed by performing, for example, laser processing on the return plate **52**. It should be noted that the communication channels **110**, **111** are both formed by substantially the same method. Therefore, the method of forming the communication channels **110**, **111** will hereinafter be described using the first communication channel **110** as an example.

[0154] The return plate processing step **S4** is provided with a connecting part forming step **S41**, an upstream opening forming step **S42**, and a downstream opening forming step **S43**.

[0155] FIG. **13** to FIG. **28** are process diagrams of the connecting part forming step **S41**. Out of FIG. **13** to FIG. **28**, FIG. **13** to FIG. **20** are each a bottom view of a portion corresponding to a portion A in FIG. **6** of the return plate **52**. Out of FIG. **13** to FIG. **28**, FIG. **21** to FIG. **28** are each a cross-sectional view of the return plate **52**. The correspondence relationship between FIG. **21** to FIG. **28** and FIG. **13** to FIG. **20** is as the cross-section pointing lines denoted by the roman numerals in FIG. **13** to FIG. **20**, respectively.

[0156] In the connecting part forming step **S41**, the connecting parts **117** are formed by a first laser irradiation step **S411** and a second laser irradiation step **S412**. Specifically, the connecting parts **117** are each formed of an outer shape part **200** (see FIG. **19** and FIG. **26**) which remains after the first laser irradiation step **S411**, and an additional-processing laser irradiation trace (a second laser irradiation trace) **201** (see FIG. **28**) which remains after the second laser irradiation step **S412**.

[0157] As shown in FIG. **13** and FIG. **21**, in the first laser irradiation step **S411**, by irradiating a surface facing to the -Z side (a lower surface) of the return plate **52** with the laser beam, the outer shape part **200** which becomes an outer shape in a plan view of the connecting part **117** is formed. Specifically, by setting the Y direction as a laser scanning direction, and setting the X direction as a pitch direction (a column direction), scanning with the laser beam is performed over a plurality of columns. Therefore, the length in the Y direction in the connecting part **117** can be adjusted in accordance with the scanning range with the laser beam. Meanwhile, the width in the X direction in the connecting part **117** can be adjusted with the number of columns of the laser irradiation in a

central column which is performed between the laser irradiation in a starting column and the laser irradiation in the last column, a laser diameter of the laser beam, an offset amount of the laser beam between columns adjacent to each other, and so on. In the first embodiment, the laser diameters in the respective columns are equivalent to each other (e.g., about 10 μm). However, the laser diameter may be changed by the columns.

[0158] In the first laser irradiation step **S411**, first, irradiation with a starting-column laser beam **L1** is performed. Specifically, by scanning a position corresponding to a $-X$ -side end portion of the connecting part **117** in the lower surface of the return plate **52** with the starting-column laser beam **L1** along the Y direction, a starting-column irradiation trace (a first laser irradiation trace) **210** shown in FIG. **14** and FIG. **22** is formed. As shown in FIG. **14**, the starting-column irradiation trace **210** extends linearly along the Y direction in the plan view. As shown in FIG. **22**, the starting-column irradiation trace **210** is recessed in a semicircular shape with respect to a lower surface of the return plate **52** when viewed from the Y direction. It should be noted that the width of the starting-column irradiation trace **210** is made slightly larger than the laser diameter since an influence of the heat of the starting-column laser beam **L1** propagates to an outside range of the laser diameter.

[0159] Incidentally, there is a possibility that a portion (hereinafter referred to as a “wrinkle”) protruding from an opening edge of the laser irradiation trace is formed after the laser irradiation. In this case, for example, an outer peripheral wrinkle **211** which protrudes from the lower surface of the return plate **52** is formed on the opening edge of the starting-column irradiation trace **210** after forming the starting-column irradiation trace **210**. The outer peripheral wrinkle **211** extends throughout the entire circumference of the starting-column irradiation trace **210** so as to surround the starting-column irradiation trace **210**.

[0160] Subsequently, as shown in FIG. **15** and FIG. **23**, by performing scanning with a second-column (central-column) laser beam **L2**, a central-column irradiation trace (the first laser irradiation trace) **215** shown in FIG. **16** and FIG. **24** is formed. As shown in FIG. **15** and FIG. **23**, the scanning with the central-column laser beam **L2** is performed while the center of the laser diameter in the central-column laser beam **L2** is provided with an offset toward the $+X$ side with respect to the center of the laser diameter in the previous column (the starting-column laser beam **L1**). In the first embodiment, an amount of the offset of the central-column laser beam **L2** is, for example, a half of the laser diameter (half pitch). However, the amount of the offset of the central-column laser beam **L2** can be changed as appropriate as long as the amount of the offset is within a range in which the starting-column irradiation trace **210** and the central-column irradiation trace **215** partially overlap each other.

[0161] The central-column irradiation trace **215** is formed so as to overlap the $+X$ -side end portion of the laser irradiation trace in the previous column (the starting-column irradiation trace **210**) to thereby form an intermediate recess **216** together with the starting-column irradiation trace **210**. As shown in FIG. **16**, the central-column irradiation trace **215** extends in parallel to the starting-column irradiation trace **210** in a state of being connected to the starting-column irradiation trace **210** in the plan view. As shown in FIG. **24**, a central-column irradiation trace **215** is recessed in a circular-arc shape with respect to the lower surface of the return plate **52** when viewed from the Y direction. It should be noted that the intermediate recess **216** is made deeper in a portion where the starting-column irradiation trace **210** overlaps the central-column irradiation trace **215** than in the starting-column irradiation trace **210**. Although when just one central-column irradiation trace **215** is provided is described in the first embodiment, the number of columns of the central-column irradiation traces **215** can be changed in accordance with the width in the X direction in the connecting part **117**.

[0162] A portion which overlaps the scanning range of the central-column laser beam **L2** out of the outer peripheral wrinkle **211** formed by the scanning with the starting-column laser beam **L1** is removed by the irradiation with the central-column laser beam **L2**. On the other hand, on the

periphery of a portion which is irradiated only with the central-column laser beam L2 out of the return plate 52, a wrinkle may newly be formed by the irradiation with the central-column laser beam L2 in some cases. As a result, the outer peripheral wrinkle 211 remains on the outer peripheral edge of the intermediate recess 216 so as to surround the periphery of the intermediate recess 216. Further, in a boundary portion between the central-column irradiation trace 215 and the starting-column irradiation trace 210 out of a bottom surface of the intermediate recess 216, there is formed a first internal wrinkle 220 which protrudes from the bottom surface of the intermediate recess 216.

[0163] Then, as shown in FIG. 17 and FIG. 25, scanning with a third-column (the last-column) laser beam L3 is performed. In the first embodiment, an amount of the offset of the last-column laser beam L3 with respect to the central-column laser beam L2 is substantially the same as, for example, the amount of the offset of the central-column laser beam L2 with respect to the starting-column laser beam L1. Thus, a last-column irradiation trace (the first laser irradiation trace) 221 shown in FIG. 18 and FIG. 26 is formed. The last-column irradiation trace 221 is formed so as to overlap the +X-side end portion of the laser irradiation trace in the previous column (the central-column irradiation trace 215) to thereby form the outer shape part 200 together with the starting-column irradiation trace 210 and the central-column irradiation trace 215. The outer shape part 200 forms an outer shape in the plan view of the connecting part 117. As shown in FIG. 18, the last-column irradiation trace 221 extends in parallel to the central-column irradiation trace 215 in a state of being connected to the central-column irradiation trace 215 in the plan view. As shown in FIG. 26, the last-column irradiation trace 221 is recessed in a circular-arc shape with respect to the lower surface of the return plate 52 when viewed from the Y direction.

[0164] A portion which overlaps the scanning range of the last-column laser beam L3 out of the outer peripheral wrinkle 211 formed by the scanning with the starting-column laser beam L1 and the central-column laser beam L2 is removed by the irradiation with the last-column laser beam L3. On the other hand, on the periphery of a portion which is irradiated only with the last-column laser beam L3 out of the return plate 52, a wrinkle may newly be formed by the irradiation with the last-column laser beam L3 in some cases. As a result, the outer peripheral wrinkle 211 remains on the outer peripheral edge of the outer shape part 200 so as to surround the periphery of the outer shape part 200. It should be noted that in a boundary portion between the central-column irradiation trace 215 and the last-column irradiation trace 221, there is formed a second internal wrinkle 222 which protrudes from the bottom surface of the outer shape part 200.

[0165] This terminates the first laser irradiation step S411.

[0166] Incidentally, there is a possibility that a burr 230 adheres to an inner surface of the outer shape part 200 after the first laser irradiation step S411. The burr 230 is apt to adhere to the inner surface along the scanning direction out of the existing irradiation trace (e.g., the starting-column irradiation trace 210) since a portion melted by the laser beam flies in all directions when, for example, forming the central-column irradiation trace 215 and the last-column irradiation trace 221. In the illustrated example, the burr 230 protrudes from the return plate 52 through an opening part of the outer shape part 200. Specifically, when the width of the recess (e.g., the connecting part 117) to be formed by the laser processing is wider than the laser diameter of a single column, it results in that two or more columns of laser irradiation traces are formed in the width direction (the X direction) of the recess so as to partially overlap the laser irradiation traces with each other. On this occasion, an area which is not irradiated redundantly with the laser beams exists in a part (the -X-side end portion) of the starting-column irradiation trace 210. The burr 230 is apt to remain in the area which is not irradiated redundantly with the laser beams.

[0167] Therefore, as shown in FIG. 19 and FIG. 27, by performing the second laser irradiation step S412, the burr 230 is mainly removed. The second laser irradiation step S412 is executed by performing scanning with an additional-processing laser beam L4 along the -X-side opening edge out of the connecting part 117 (the outer shape part 200). Specifically, the additional-processing

laser beam **L4** is set to have substantially the same laser diameter as that of the laser beam used in the first laser irradiation step **S411**. However, the laser diameter may be changed between the first laser irradiation step **S411** and the second laser irradiation step **S412**.

[0168] The second laser irradiation step **S412** is performed at the irradiation position in the X direction in the additional-processing laser beam **L4** while making the center of the laser diameter in the additional-processing laser beam **L4** coincide with the center of the starting-column laser beam **L1**. Further, the scanning range in the Y direction in the additional-processing laser beam **L4** is substantially the same as the scanning range of the laser beam in the starting column. When performing the second laser irradiation step **S412** in such a condition, the irradiation with the additional-processing laser beam **L4** is performed along a scan trajectory of the starting-column laser beam **L1**. Thus, as shown in FIG. **20** and FIG. **28**, the burr **230** is removed. Further, at a place where the starting-column irradiation trace **210** is present out of the outer shape part **200**, the additional-processing laser irradiation trace **201** is formed by the additional-processing laser beam **L4**. The additional-processing laser irradiation trace **201** forms the connecting part **117** together with the outer shape part **200**. In the connecting part **117**, it results in that the outer peripheral wrinkle **211** remains on the opening edge on the one hand, and an internal wrinkle **225** remains on the bottom surface on the other hand. The internal wrinkle **225** is a wrinkle which is formed in a boundary portion of the additional-processing laser irradiation trace **201** on the bottom surface of the outer shape part **200**.

[0169] Subsequently, in the upstream opening forming step **S42**, laser processing, for example, is applied to a formation region of the upstream opening **115** in the bottom surface of the connecting part **117** so as to penetrate the return plate **52**. Thus, the upstream opening **115** is formed.

[0170] Regarding the downstream opening forming step **S43**, laser processing, for example, is applied to a formation region of the downstream opening **116** in the bottom surface of the connecting part **117** so as to penetrate the return plate **52**. Thus, the downstream opening **116** is formed. It should be noted that it may be arranged that the internal wrinkle **225** is removed when forming the upstream opening **115** and the downstream opening **116** with the laser processing.

[0171] This completes the return plate processing step **S4** (the state in FIG. **12**).

[0172] In the first embodiment, the return plate **52** is bonded to the chip modules **51A**, **51B**, and then the communication channels **110**, **111** are formed, but this configuration is not a limitation. It is possible to provide the communication channels **110**, **111** to the return plate **52** in advance, and then bond the return plate **52** to the chip modules **51A**, **51B**. Further, there is described the method of performing the first laser irradiation step **S411** and the second laser irradiation step **S412** in a row in the first embodiment, but this configuration is not a limitation. For example, it is possible to perform the first laser irradiation step **S411**, then perform the upstream opening forming step **S42** and the downstream opening forming step **S43**, and then perform the second laser irradiation step **S412** at the end of the return plate processing step **S4**.

[0173] In the protective film forming step **S5**, the protective films **125** are formed on the inner surface of the common ink chamber **80**, the inner surfaces of the slits **81**, the inner surfaces of the ejection channels **71**, the inner surfaces of the communication channels **110**, **111**, the inner surfaces of the connecting channels **121**, **122**, and the inner surface of the manifold **123**. The protective films **125** are formed by depositing a para-xylylene resin material using, for example, a chemical vapor deposition method (CVD).

[0174] In the nozzle plate stacking step **S6**, the nozzle plate **53** is bonded to the lower surface of the return plate **52**.

[0175] Due to the steps described hereinabove, the head chip **50** is completed. It should be noted that when manufacturing the head chips **50** wafer by wafer, substantially the same step as the module forming step **S1** described above is performed on an actuator plate wafer, a cover plate wafer, and a back plate wafer to thereby form a stacked body of the wafers. Subsequently, by segmentalizing the stacked body of the wafers, the plurality of chip modules **51A**, **51B** is taken out.

Subsequently, by performing the module stacking step S2 and subsequent steps on the chip modules 51A, 51B thus taken out from the stacked body of the wafers, the head chips 50 are completed.

[0176] As described above, the head chip 50 according to the first embodiment is provided with the return plate (the flow channel member) 52 and the nozzle plate (the bonding member) 53, wherein the return plate 52 has the connecting part (the recess) 117 formed of the outer shape part 200 (the irradiation traces 210, 215, and 221), the ink is circulated through the connecting part 117, and the nozzle plate 53 is bonded to the return plate 52 so as to close the connecting part 117. In the return plate 52, the additional-processing laser irradiation trace 201 extending along the opening edge of the connecting part 117 is formed in the connecting part 117.

[0177] According to this configuration, since the additional-processing laser irradiation trace 201 is formed along the opening edge of the connecting part 117, it is possible to reduce the unwanted processing residuals (the burr 230 and so on) remaining on the opening edge of the connecting part 117 (the outer shape part 200) after the first laser irradiation step S411. Thus, it is possible to prevent the unwanted processing residuals from being jammed when bonding the return plate 52 and the nozzle plate 53 to each other. As a result, it is possible to prevent the leakage of the ink between the return plate 52 and the nozzle plate 53, and at the same time, it is possible to ensure the bonding strength between the return plate 52 and the nozzle plate 53. Further, by reducing the unwanted processing residuals in the connecting part 117, it is possible to suppress the unevenness in the connecting part 117. Thus, it is possible to reduce the circulation resistance of the ink which circulates in the connecting part 117 to thereby smoothly circulate the ink in the connecting part 117.

[0178] In the head chip 50 according to the first embodiment, the communication channels 110, 111 are each provided with the connecting part 117 which opens only on the surface facing to the -Z side (the surface facing to the bonded member side in the thickness direction) out of the return plate 52, and the upstream opening 115 and the downstream opening 116 (penetrating parts) which open on the bottom surface of the connecting part 117, and at the same time, penetrate the return plate 52 in the Z direction. The additional-processing laser irradiation trace 201 extends along at least the opening edge of the connecting part 117.

[0179] According to this configuration, there is a possibility that the unwanted processing residuals remaining on the opening edge of the connecting part 117 are exposed on a bonding surface of the return plate 52 with the nozzle plate 53. Therefore, by forming the additional-processing laser irradiation trace 201 along at least the opening edge of the connecting part 117, it is possible to prevent the unwanted processing residuals from being exposed on the bonding surface of the return plate 52 with the nozzle plate 53.

[0180] In the head chip 50 according to the first embodiment, there is adopted the configuration in which the plurality of columns of irradiation traces 210, 215, and 221 constituting the outer shape part 200 is formed in the X direction, and the additional-processing laser irradiation trace 201 extends in the Y direction along the starting-column irradiation trace (an outside laser irradiation trace) 210.

[0181] According to this configuration, the unwanted processing residuals such as the burr 230 fly in all directions due to the laser processing, and are therefore apt to adhere to the inner surfaces of the existing laser irradiation traces when performing the laser processing in the second and subsequent columns. In particular, the starting-column irradiation trace 210 forms the opening part of the connecting part 117 along the extending direction, and is therefore apt to lead to the leakage of the ink and the bonding failure described above when the unwanted processing residuals remain on the inner surfaces of the starting-column irradiation trace 210. Therefore, by forming the additional-processing laser irradiation trace 201 along the starting-column irradiation trace 210 out of the plurality of irradiation traces 210, 215, and 221, it is possible to effectively reduce the unwanted processing residuals remaining along the extending direction of the starting-column

irradiation trace **210**.

[0182] In the head chip **50** according to the first embodiment, there is adopted the configuration in which the connecting part **117** is formed to have a rectangular shape having the longitudinal direction set to the Y direction and the transverse direction set to the X direction when viewed from the Z direction.

[0183] According to this configuration, since the longitudinal direction of the connecting part **117** and the extending direction of the additional-processing irradiation trace **201** coincide with each other, the unwanted processing residuals can be reduced along the longitudinal direction of the connecting part **117**.

[0184] In the head chip **50** according to the first embodiment, the communication channels **110**, **111** each include the upstream opening (first penetrating part) **115** and the downstream opening (second penetrating part) **116** which is disposed at a distance from the upstream opening **115** in the Y direction, and is communicated with the upstream opening **115** through the connecting part **117**. The chip modules (chip main bodies) **51A**, **51B**, which are provided with the ejection channels (pressure chambers) **71** communicated with the communication channels **110**, **111** through the upstream openings **115**, and the connecting channels (circulation channels) **121**, **122** communicated with the communication channels **110**, **111** through the downstream openings **116**, are bonded to the return plate **52** at an opposite side to the nozzle plate **53**, and the nozzle plate **53** is provided with the nozzle holes (jet holes) **131**, **132** for ejecting the ink in the communication channels **110**, **111** to the outside.

[0185] According to this configuration, the flow channel member has a function as the return plate **52** which is communicated with the nozzle holes **131**, **132** and circulates the ink between the ejection channels **71** and the connecting channels **121**, **122**. In this case, in the head chip **50** according to the first embodiment, since it is possible to suppress the unevenness in the communication channels **110**, **111**, it is possible to smoothly perform the circulation of the ink between the nozzle holes **131**, **132**, the ejection channels **71**, and the connecting channels **121**, **122** through the communication channels **110**, **111**.

[0186] Since the inkjet head **5** and the printer **1** according to the first embodiment are each provided with the head chip **50** described above, it is possible to provide the inkjet head **5** and the printer **1** which are excellent in reliability.

Second Embodiment

[0187] FIG. **29** to FIG. **32** are process diagrams of a second laser irradiation step **S412** related to a second embodiment. Out of FIG. **29** to FIG. **32**, FIG. **29** and FIG. **30** are each a bottom view of a portion corresponding to the portion A in FIG. **6**. Out of FIG. **29** to FIG. **32**, FIG. **31** and FIG. **32** are each a cross-sectional view of the return plate **52**.

[0188] The second embodiment is different from the first embodiment described above in the point that the second laser irradiation step **S412** is performed in the state in which the center of the additional-processing laser beam **L4** is disposed at a position shifted in the X direction from the center of the starting-column laser beam **L1**. Specifically, in the second laser irradiation step **S412** related to the second embodiment, irradiation with the additional-processing laser beam **L4** is performed so that the additional-processing laser irradiation trace **201** is formed at the inner side (the +X side) in the X direction with respect to the opening edge of the connecting part **117** (the starting-column irradiation trace **210**). In this case, it is preferable to perform the irradiation so that the center of the laser diameter in the additional-processing laser beam **L4** is located between the center of the laser diameter in the starting-column laser beam **L1** and the center of the laser diameter in the central-column laser beam **L2**. However, the center of the laser diameter in the additional-processing laser beam **L4** can be changed to a position as appropriate as long as the heat of at least the additional-processing laser beam **L4** propagates to the burr **230** at that position.

[0189] Then, due to the influence of the heat which propagates to the periphery of the additional-processing laser beam **L4**, the burr **230** is removed. Specifically, when forming the additional-

processing laser irradiation trace **201**, the burr **230** protruding from the opening edge of the connecting part **117** and so on can effectively be removed.

[0190] Further, the additional-processing laser irradiation trace **201** is formed so that the center in the X direction is located in a portion located between the center in the X direction in the starting-column irradiation trace **210** and the center in the X direction in the central-column irradiation trace **215**. In other words, the center (the deepest portion) in the X direction in the additional-processing laser irradiation trace **201** is disposed at a position shifted from the centers (the deepest portions) in the X direction in the irradiation traces **210**, **215** adjacent to each other.

[0191] In the head chip **50** according to the second embodiment, there is adopted the configuration in which the center in the X direction in the additional-processing laser irradiation trace **201** is located at the inner side in the X direction from the center in the X direction in the starting-column irradiation trace **210**.

[0192] According to this configuration, by disposing the center in the X direction in the additional-processing laser irradiation trace **201** at the inner side in the X direction from the center in the X direction in the starting-column irradiation trace **210**, it is possible to remove the burr **230** due to the influence of the heat propagating to the periphery of the additional-processing laser beam **L4**. In this case, the influence of the heat propagating to the opening edge of the outer shape part **200** (the connecting part **117**) can be suppressed compared to when performing the scanning with the additional-processing laser beam **L4** on the same trajectory as the scan trajectory of the starting-column laser beam **L1**. Thus, it is possible to prevent the outer peripheral wrinkle **211** from being enlarged by the additional-processing laser beam **L4**. Therefore, it is possible to prevent the outer peripheral wrinkle **211** from being jammed as the unwanted processing residuals when bonding the return plate **52** and the nozzle plate **53** to each other. As a result, it is possible to prevent the leakage of the ink between the return plate **52** and the nozzle plate **53**, and at the same time, it is possible to ensure the bonding strength between the return plate **52** and the nozzle plate **53**.

[0193] Moreover, since the center in the X direction is shifted between the additional-processing laser irradiation trace **201**, and the starting-column irradiation trace **210** and the central-column irradiation trace **215**, it is also possible to prevent the inner peripheral wrinkle from being enlarged. As a result, since the unevenness in the connecting part **117** can be suppressed, it is possible to reduce the circulation resistance of the ink which circulates in the communication channels **110**, **111** to thereby smoothly circulate the ink in the communication channels **110**, **111**.

Third Embodiment

[0194] A third embodiment is different from the embodiments described above in the point that the laser irradiation for additional processing is performed with respect to the method of forming the upstream opening **115** which forms the penetrating part. FIG. **33** is a flowchart illustrating a method of manufacturing a head chip **50** according to the third embodiment. FIG. **34** to FIG. **49** are process diagrams of the upstream opening forming step **S42**. Out of FIG. **34** to FIG. **49**, FIG. **34** to FIG. **41** are each a bottom view of a portion corresponding to the portion A in FIG. **6** of the return plate **52**. Out of FIG. **34** to FIG. **49**, FIG. **42** to FIG. **49** are each a cross-sectional view of the return plate **52**. The correspondence relationship between FIG. **42** to FIG. **49** and FIG. **34** to FIG. **41** is as the cross-section pointing lines denoted by the roman numerals in FIG. **34** to FIG. **41**, respectively.

[0195] As shown in FIG. **33**, the upstream opening forming step **S42** is provided with the first laser irradiation step **S421** and the second laser irradiation step **S422**. In the following description, regarding the constituents substantially the same as those of the first embodiment, the description will be omitted as appropriate.

[0196] In the first laser irradiation step **S421**, the bottom surface of the connecting part **117** out of the return plate **52** is irradiated with the laser beam to thereby form the outer shape in the plan view of the upstream opening **115**. Specifically, by setting the Y direction as a laser scanning direction, and setting the X direction as a pitch direction (a column direction), scanning with the laser beam is performed over a plurality of columns. As shown in FIG. **34** and FIG. **42**, in the first laser

irradiation step **S421**, irradiation with the starting-column laser beam **L1** is performed. Specifically, by scanning a position corresponding to the $-X$ -side end portion of the upstream opening **115** in the bottom surface of the connecting part **117** with the laser beam along the Y direction, a starting-column irradiation trace (the first laser irradiation trace) **300** shown in FIG. **35** and FIG. **43** is formed. As shown in FIG. **35**, the starting-column irradiation trace **300** extends linearly along the Y direction in the plan view. As shown in FIG. **43**, the starting-column irradiation trace **300** penetrates the return plate **52** via the connecting part **117**. It should be noted that after forming the starting-column irradiation trace **300**, an outer peripheral wrinkle **301** which protrudes from the bottom surface of the connecting part **117** is present on the opening edge of the starting-column irradiation trace **300**.

[0197] Subsequently, as shown in FIG. **36** and FIG. **44**, scanning with a second-column (the central-column) laser beam **L2** is performed. The scanning with the central-column laser beam **L2** is performed while the center of the laser diameter is provided with an offset toward the $+X$ side with respect to the center of the laser diameter in the previous column (the starting-column laser beam **L1**). Thus, a central-column irradiation trace **305** shown in FIG. **37** and FIG. **45** is formed. The central-column irradiation trace **305** is formed so as to overlap the $+X$ -side end portion of the laser irradiation trace in the previous column (the starting-column irradiation trace **300**) to thereby form an intermediate recess **306** together with the starting-column irradiation trace **300**. As shown in FIG. **37**, the central-column irradiation trace **305** extends in parallel to the starting-column irradiation trace **300** in a state of being connected to the starting-column irradiation trace **300** in the plan view. As shown in FIG. **45**, the central-column irradiation trace **305** penetrates the return plate **52** together with the starting-column irradiation trace **300**. It should be noted that although when just one central-column irradiation trace **305** is provided is described in the third embodiment, the number of columns of the central-column irradiation traces **305** can be changed in accordance with the width in the X direction in the upstream opening **115**.

[0198] Then, as shown in FIG. **38** and FIG. **46**, scanning with the last-column (a third-column) laser beam **L3** is performed. In the third embodiment, an amount of the offset of the last-column laser beam **L3** is substantially the same as, for example, the amount of the offset of the central-column laser beam **L2** with respect to the starting-column laser beam **L1**. Thus, a last-column irradiation trace **311** shown in FIG. **39** and FIG. **47** is formed. The last-column irradiation trace **311** is formed so as to overlap the $+X$ -side end portion of the laser irradiation trace in the previous column (the central-column irradiation trace **305**) to thereby penetrate the return plate **52** together with the starting-column irradiation trace **300** and the central-column irradiation trace **305**. The last-column irradiation trace **311** extends in parallel to the central-column irradiation trace **305** in a state of being connected to the central-column irradiation trace **305**.

[0199] This terminates the first laser irradiation step **S421**. In other words, in the first laser irradiation step **S421**, an outer shape part **315** having an outer shape in the plan view equivalent to that of the upstream opening **115** is formed.

[0200] After the first laser irradiation step **S421**, there is a possibility that the portion melted by, for example, the central-column laser beam **L2** in the formation of, for example, the central-column irradiation trace **305** adheres to the inner surface of the starting-column irradiation trace **300** as a burr **320** in the inner surface of the outer shape part **315** (the upstream opening **115**).

[0201] Therefore, as shown in FIG. **40** and FIG. **48**, by performing the second laser irradiation step **S422**, the burr **320** is mainly removed. The second laser irradiation step **S422** is executed by performing irradiation with the additional-processing laser beam **L4** along the $-X$ -side opening edge out of the outer shape part **315** (the upstream opening **115**). Specifically, the center of the laser diameter in the additional-processing laser beam **L4** coincides with the center of the starting-column laser beam **L1**. Meanwhile, the scanning range in the Y direction in the additional-processing laser beam **L4** is substantially the same as the scanning range of the starting-column laser beam **L1**. When performing the second laser irradiation step **S422** in such a condition, the

irradiation with the additional-processing laser beam is performed along the scan trajectory of the starting-column laser beam L1. Thus, as shown in FIG. 41 and FIG. 49, the burr 320 is removed. Further, at a place where the starting-column irradiation trace 300 is present out of the outer shape part 315, an additional-processing laser irradiation trace (the second laser irradiation trace) 321 is formed by the additional-processing laser beam L4. The additional-processing laser irradiation trace 321 forms the upstream opening 115 together with the laser irradiation traces 300, 305, and 311.

[0202] In the second embodiment, by executing the first laser irradiation step S421 and the second laser irradiation step S422 on the upstream opening 115 as the penetrating part, substantially the same functions and advantages as those of the first embodiment described above are exerted.

[0203] Further, by removing the burr 320 in advance before bonding the return plate 52 and the nozzle plate 53 to each other, it is possible to prevent the burr 320 from dropping and then being retained in the ink after bonding the return plate 52 and the nozzle plate 53 to each other. It should be noted that in the third embodiment, there is described when the method of manufacturing the head chip 50 according to the present disclosure is adopted in the upstream opening 115, but it is also possible to apply the method of manufacturing the head chip 50 according to the present disclosure to the processing of the downstream opening 116.

Fourth Embodiment

[0204] FIG. 50 to FIG. 53 are process diagrams of a second laser irradiation step S422 related to a fourth embodiment. Out of FIG. 50 to FIG. 53, FIG. 50 and FIG. 51 are each a bottom view of a portion corresponding to the portion A in FIG. 6. Out of FIG. 50 to FIG. 53, FIG. 52 and FIG. 53 are each a cross-sectional view of the return plate 52.

[0205] The fourth embodiment is different from the third embodiment described above in the point that the second laser irradiation step S422 is performed in the state in which the center of the additional-processing laser beam L4 is disposed at a position shifted in the X direction from the center of the starting-column laser beam L1. Specifically, in the second laser irradiation step S422 related to the fourth embodiment, irradiation with the additional-processing laser beam L4 is performed on the portion located at an inner side in the X direction with respect to the opening edge of the upstream opening 115. In this case, it is preferable to perform the irradiation with the additional-processing laser beam L4 so that the center of the laser diameter is located between the center of the laser diameter in the starting-column laser beam L1 and the center of the laser diameter in the central-column laser beam L2. However, the center of the laser diameter in the additional-processing laser beam L4 can be changed to a position as appropriate as long as the heat of at least the additional-processing laser beam L4 propagates to the burr 320 at that position.

[0206] Then, due to the influence of the heat which propagates to the periphery of the additional-processing laser beam L4, the burr 320 is removed. Specifically, the burr 320 protruding from the opening edge of the upstream opening 115 and so on can effectively be removed by the additional-processing laser beam L4. Further, in the fourth embodiment, the additional-processing laser irradiation trace 321, which is formed due to the influence of the heat of the additional-processing laser beam L4, remains in a portion located between the starting-column irradiation trace 300 and the central-column irradiation trace 305 on a surface facing to the +X side and a surface facing to the inner side in the Y direction out of the inner surfaces of the upstream opening 115.

[0207] Also in the fourth embodiment, similarly to the second embodiment described above, it is possible to prevent the outer peripheral wrinkle 301 remaining on the opening edge of the upstream opening 115 from being enlarged by the additional-processing laser irradiation trace 321.

Therefore, it is possible to suppress the unevenness in the communication channels 110, 111. Thus, it is possible to reduce the circulation resistance of the ink which circulates in the communication channels 110, 111 to thereby smoothly circulate the ink in the communication channels 110, 111.

Other Modified Examples

[0208] It should be noted that the scope of the present disclosure is not limited to the embodiments

described above, but a variety of modifications can be applied within the scope or the spirit of the present disclosure.

[0209] For example, in the embodiments described above, the description is presented citing the inkjet printer **1** as an example of the liquid jet recording apparatus, but the liquid jet recording apparatus is not limited to the printer. For example, a facsimile machine, an on-demand printing machine, and so on can also be adopted.

[0210] In the embodiments described above, the description is presented citing the configuration (a so-called shuttle machine) in which the inkjet heads move with respect to the recording target medium when performing printing as an example, but this configuration is not a limitation. The configuration related to the present disclosure can be adopted as the configuration (a so-called stationary head machine) in which the recording target medium is moved with respect to the inkjet heads in the state in which the inkjet heads are fixed.

[0211] In the embodiments described above, there is explained when the recording target medium P is paper, but this configuration is not a limitation. The recording target medium P is not limited to paper, but can also be a metal material or a resin material, and can also be food or the like.

[0212] In the embodiments described above, there is explained the configuration in which the liquid jet heads are installed in the liquid jet recording apparatus, but this configuration is not a limitation. Specifically, the liquid to be jetted from the liquid jet heads is not limited to what is landed on the recording target medium, but can also be, for example, a medical solution to be blended during a dispensing process, a food additive such as seasoning or a spice to be added to food, or fragrance to be sprayed in the air.

[0213] In the embodiments described above, there is explained the configuration in which the Z direction coincides with the gravitational direction, but this configuration is not a limitation, and it is also possible to set the Z direction to a direction along the horizontal direction.

[0214] In the embodiments described above, there is explained the configuration (so-called pulling-shoot) of deforming the actuator plate in the direction of increasing the volume of the ejection channel due to the application of the voltage, and then restoring the actuator plate to thereby eject the ink, but this configuration is not a limitation. It is possible for the head chip according to the present disclosure to be provided with a configuration (so-called pushing-shoot) in which the ink is ejected by deforming the actuator plate in a direction of reducing the volume of the ejection channel due to the application of the voltage. When performing the pushing-shoot, the actuator plate deforms so as to bulge toward the inside of the ejection channel due to the application of the drive voltage. Thus, the volume in the ejection channel decreases to increase the pressure in the ejection channel, and thus, the ink located in the ejection channel is ejected outside through the nozzle hole. When setting the drive voltage to zero, the actuator plate is restored. As a result, the volume in the ejection channel is restored.

[0215] In the embodiment described above, there is explained the configuration in which the chip modules **51A**, **51B** are overlapped with each other, but this configuration is not a limitation. It is possible to configure the head chip **50** only with the first chip module **51A**.

[0216] In the embodiments described above, the description is presented citing the head chip of the edge-shoot type as an example, but this configuration is not a limitation. For example, it is also possible to apply the present disclosure to the head chip **50** of a so-called side-shoot type for ejecting the ink from a central portion in the extending direction in the ejection channel **71**. Further, the head chip may be of a so-called roof-shoot type in which a direction of the pressure applied to the ink and the ejection direction of the ink are made the same as each other. In the case of the roof-shoot type, a portion which faces the pressure chamber retaining the ink, and which deforms for expanding or contracting the pressure chamber, functions as a drive unit.

[0217] The configuration in which the scanning direction with the laser beam in the first laser irradiation step and the scanning direction with the laser beam in the second laser irradiation step coincide with each other is explained in the embodiments described above, but this configuration is

not a limitation. The scanning direction with the laser beam in the first laser irradiation step and the scanning direction with the laser beam in the second laser irradiation step may cross each other. [0218] In the embodiments described above, there is described when the laser irradiation in the additional processing is only performed along the opening edge formed of the starting-column irradiation trace, but this configuration is not a limitation. The laser irradiation in the additional processing may be performed over the entire circumference of the opening edge of the first laser irradiation trace.

[0219] In the embodiments described above, there is described when only the connecting part **117** is formed adopting the method of manufacturing the head chip **50** according to the present disclosure in the first embodiment, and only the upstream opening **115** (or the downstream opening **116**) is formed adopting the method of manufacturing the head chip **50** according to the present disclosure in the third embodiment, but this configuration is not a limitation. The method of manufacturing the head chip **50** according to the present disclosure may be adopted with respect to both the connecting part **117** and the variety of openings **115**, **116**.

[0220] In the embodiments described above, there is described the configuration in which the recess related to the present disclosure is provided with the connecting part **117** as the non-penetrating part, and the openings **115**, **116** as the penetrating parts, but this configuration is not a limitation. The recess may be formed only of the non-penetrating part, or may also be formed only of the penetrating part.

[0221] In the embodiments described above, the return plate **52** is described as the flow channel member, and the nozzle plate **53** is described as the bonded member, but this configuration is not a limitation. The head chip according to the present disclosure can be applied to a variety of members which configures the head chip in which a member provided with a recess as a flow channel is defined as the flow channel member, and a member to be bonded to the flow channel member is defined as the bonded member.

[0222] Besides the above, it is arbitrarily possible to replace the constituents in the embodiments described above with known constituents within the scope or the spirit of the present disclosure, and it is also possible to arbitrarily combine the modified examples described above with each other.

Claims

1. A head chip comprising: a flow channel member which includes a recess formed of a first laser irradiation trace, and in which a liquid circulates in the recess; and a bonded member which is bonded to the flow channel member so as to close the recess, wherein the flow channel member is provided with a second laser irradiation trace extending along an opening edge of the recess.
2. The head chip according to claim 1, wherein the recess includes a non-penetrating part configured to open only on a surface of the flow channel member, the surface facing to the bonded member side in a thickness direction, and a penetrating part which is configured to open on a bottom surface of the non-penetrating part, and penetrates the flow channel member in the thickness direction, and the second laser irradiation trace extends along at least an opening edge of the non-penetrating part.
3. The head chip according to claim 1, wherein the first laser irradiation trace extends in a first direction crossing a thickness direction of the flow channel member, and a plurality of first laser irradiation traces each identical to the first laser irradiation trace is formed in a second direction crossing the first direction when viewed from the thickness direction, and the second laser irradiation trace extends in the first direction along an outer side laser irradiation trace which is located at an outermost side in the second direction in the plurality of first laser irradiation traces.
4. The head chip according to claim 3, wherein a center in the second direction in the second laser irradiation trace is located at an inner side in the second direction from a center in the second

direction in the outer side laser irradiation trace.

5. The head chip according to claim 3, wherein the recess is formed to have a rectangular shape having a longitudinal direction set to the first direction and a transverse direction set to the second direction when viewed from the thickness direction.

6. The head chip according to claim 3, wherein the recess includes a non-penetrating part configured to open only on a surface of the flow channel member, the surface facing to the bonded member side in a thickness direction, and a penetrating part which is configured to open on a bottom surface of the non-penetrating part, and penetrates the flow channel member in the thickness direction, and the penetrating part includes a first penetrating part, and a second penetrating part which is disposed at a distance in the first direction with respect to the first penetrating part, and is communicated with the first penetrating part through the non-penetrating part, a chip main body provided with a pressure chamber communicated with the recess through the first penetrating part and a circulation channel communicated with the recess through the second penetrating part is bonded to the flow channel member at an opposite side to the bonded member in the thickness direction, and the bonded member is provided with a jet hole configured to jet a liquid in the recess to an outside.

7. A liquid jet head comprising the head chip according to claim 1.

8. A liquid jet recording apparatus comprising the liquid jet head according to claim 7.

9. A method of manufacturing a head chip including a flow channel member provided with a recess through which a liquid circulates, and a bonded member which is bonded to the flow channel member so as to close the recess, the method comprising: a first laser irradiation step of irradiating a bonding surface of the flow channel member to the bonded member with a first laser beam to thereby form an outer shape of the recess with a first laser irradiation trace; and a second laser irradiation step of irradiating the bonding surface with a second laser beam along an opening edge of the recess to thereby form a second laser irradiation trace along the opening edge of the recess.
